№ T5

$\xi \sim R[\theta, 2\theta]$, $M\xi = \frac{\theta+2\theta}{2} = \frac{3}{2}\theta$, $D\xi = \frac{(2\theta-\theta)^2}{12} = \frac{\theta^2}{12}$

а) Оценка методом Моментов: $M\xi = \bar{x}$; $\tilde{\theta}_1 = \frac{2}{3}\bar{x}$

Оценка методом макс. правдоподобия: $L(\theta) = \prod_{i=1}^{n} p(x, \theta) = \frac{1}{\theta^n}$

$x_{min} \geq \theta$, $x_{max} \leq 2\theta \Rightarrow \frac{x_{max}}{2} \leq \theta \leq x_{min}$

$\Rightarrow \sup L(\theta) = L\left(\frac{x_{max}}{2}\right) \Rightarrow \tilde{\theta}_2 = \frac{x_{max}}{2}$

б) Проверка на несмещенность:

$M\tilde{\theta}_1 = M\left[\frac{2}{3}\frac{1}{n}\sum_{i=1}^{n} x_i\right] = \frac{2}{3n} n M\xi = \frac{2}{3}\cdot\frac{3}{2}\theta = \theta \Rightarrow$ Оценка Несмещенная

$M\tilde{\theta}_2 = M\left[\frac{x_{max}}{2}\right] = \frac{1}{2}Mx_{max}$, $P(x_{max} < y) = \Phi(y) = F^n(y)$

$\varphi(y) = nF^{n-1}(y)\,p(y) = \frac{n}{\theta}\cdot\left(\frac{y-\theta}{\theta}\right)^{n-1}$

$\Rightarrow M\tilde{\theta}_2 = \frac{1}{2}\int_{-\infty}^{+\infty} y\frac{n}{\theta^n}(y-\theta)^{n-1}dy = \frac{n}{2\theta^n}\int_{\theta}^{2\theta} y(y-\theta)^{n-1}dy =$

$(y-\theta = t)$

$= \frac{1}{2}\frac{n}{\theta^n}\int_0^{\theta}(t+\theta)t^{n-1}dt = \frac{1}{2}\frac{n}{\theta^n}\left(\frac{\theta^{n+1}}{n+1} + \frac{\theta^{n+1}}{n}\right) = \frac{1}{2}\frac{\theta n(2n+1)}{n(n+1)} =$

$= \theta\frac{2n+1}{2n+2}$ — Смещенная $\Rightarrow$ Исправл. оценка

$\tilde{\theta}_2' = \frac{2n+2}{2n+1}\cdot\frac{x_{max}}{2}$

будет Несмещенная

Проверка Состоятельность

$D\tilde{\theta}_1 = \left(\frac{2}{3}\right)^2\frac{1}{n^2}nD\xi = \frac{4}{9n}\frac{\theta^2}{12} = \frac{\theta^2}{27n} \xrightarrow[n\to\infty]{} 0 \Rightarrow$ Оценка Состоятельна

$M\tilde{\theta}_2^2 = \frac{1}{4}Mx_{max}^2 = \frac{1}{4}\int_{\theta}^{2\theta} y^2(y-\theta)^{n-1}\frac{n}{\theta^n}dy = \frac{1}{4}\int_0^{\theta}(t+\theta)^2\frac{n}{\theta^n}t^{n-1}dt =$

$= \frac{n}{4\theta^n}\int_0^{\theta}(t^{n+1} + 2\theta t^n + \theta^2 t^{n-1})dt = \frac{1}{4}\frac{n}{\theta^n}\left(\frac{\theta^{n+2}}{n+2} + \frac{2\theta^{n+2}}{n+1} + \frac{\theta^{n+2}}{n}\right) =$

$= \frac{1}{4}\theta^2\frac{\cancel{n}(n^2+n+2n+4n+n^2+2n+n+2)}{\cancel{n}(n+1)(n+2)} = \frac{\theta^2}{4}\frac{2n^2+4n+1}{(n+1)(n+2)}$

$\Rightarrow D\tilde{\theta}_2^2 = M\tilde{\theta}_2^2 - (M\tilde{\theta}_2)^2 = \frac{1}{2}\theta^2 \frac{2n^2+4n+1}{(n+1)(n+2)} - \theta^2 \frac{4n^2+4n+1}{4(n+1)^2} =$

$= \theta^2 \left(\frac{\cancel{4n^3+8n^2+9n+n^2+8n+2-4n^3-4n^2-n-8n^2-8n-2}}{4(n+1)^2(n+2)} \right) =$

$= \theta^2 \frac{n}{4(n+1)^2(n+2)}$ — $D\tilde{\theta}_2$

$\tilde{\theta}_2' = \frac{2n+2}{2n+1}\tilde{\theta}_2 \Rightarrow D\tilde{\theta}_2' = \left(\frac{2n+2}{2n+1}\right)^2 D\tilde{\theta}_2 =$

$= \underbrace{\left(\frac{2n+2}{2n+1}\right)^2}_{\xrightarrow[n\to\infty]{} 1} \frac{n}{4(n+1)^2(n+2)}\theta^2 \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_2'$ Состоятельна по дост. условию

т.к. несмещена

c) - Проверка эффективности.

$D\tilde{\theta}_1 = \frac{\theta^2}{27n}$; $D\tilde{\theta}_2' = \left(\frac{2n+2}{2n+1}\right)^2 \frac{n}{4(n+1)^2(n+2)}\theta^2 = \frac{n}{(2n+1)^2(n+2)}\theta^2$

$4(n+1)^2/(2n+1)^2$

$\tilde{\theta}_1$ $\tilde{\theta}_2'$

$\frac{\theta^2}{27n} \times \frac{n}{(2n+1)^2(n+2)}\theta^2 \Rightarrow (2n+1)^2(n+2) \dots 27n^2$

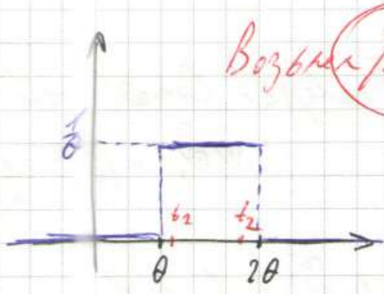
$4n^3+12n^2+9n+2 \dots 27n^2$

$4n^3-15n^2+9n+2 \geq 0 \Rightarrow n \geq 3 \Rightarrow D\tilde{\theta}_1 > D\tilde{\theta}_2' \Rightarrow$

$\Rightarrow \tilde{\theta}_2'$ более эффективна (при $n \geq 3$)

d) - Точный довер. интервал:

Возьмем $\beta = 0.95$



$t_1 = q_{\frac{1-\beta}{2}} = \theta + \frac{\theta}{20}\cdot\frac{1}{2} = \frac{41}{40}\theta$

$t_2 = q_{\frac{1+\beta}{2}} = 2\theta - \frac{\theta}{40} = \frac{79}{40}\theta$.

$F(x) = \frac{x-\theta}{\theta}$, пусть $\gamma = \frac{X_{max}}{\theta} \Rightarrow$ на $[1;2]$ $t_1 = q_{\frac{1-\beta}{2}}$; $t_2 = q_{\frac{1+\beta}{2}}$

$\Rightarrow P(\gamma \le y) = (y-1)^n$ $y \in [1;2] \Rightarrow P(t_1 \le \gamma \le t_2) = \beta$

$f(\vec{x}_n, \theta) = \frac{X_{max}}{\theta} = y_{max} \sim \varphi(t)$ $\varphi(t) = n(t-1)^{n-1}$; $f(\vec{x}_n, \theta) = \frac{X_{max}}{\theta} \sim \varphi(t)$

$\Rightarrow P\left(\frac{x_{max}}{t_2} < \theta < \frac{x_{max}}{t_2}\right) = \beta$

$\int_1^{t_1} \varphi(t)\,dt = 0.025 \Rightarrow F(t_1) - F(1) = 0.025$

$\Rightarrow t_1 = 1 + (0.025)^{-n}$

Итого также $t_2 = (1 + (0.975)^{-n}$

$\Rightarrow$ Точный довер. интервал: $\left(\frac{x_{max}}{t_2} ; \frac{x_{max}}{t_1}\right)$

e) Асимптотический доверит. интервал для параметра θ

Возьмем оценку из ОМП $\tilde{\theta}_2' = \frac{2n+2}{2n+1} \cdot \frac{x_{max}}{2}$

$$\Rightarrow \int_{-\infty}^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right) = \int_{-\infty}^{+\infty} d\left(\frac{-\theta-(x-\theta)}{\theta^2}\right) = -\int_{\theta}^{2\theta} \frac{1}{\theta^2}\,dx + 0 = \frac{\partial}{\partial\theta}\int_{\infty}^{+\infty} dF$$

А вот все, все, модель не явл. регулярной $\Rightarrow$ она не является сильно регулярной $\Rightarrow$ ОМП нам не подходит :(

Возьмем $\tilde{\theta}_1$. Св-во разобрано, т.к. выполн. усл. простейшей ЦПТ

$$\frac{\tilde{\alpha}_k - \alpha_k}{\sqrt{\alpha_{2k} - \alpha_k^2}}\sqrt{n} \to N(0,1), \quad \tilde{\theta}_1 = g(\tilde{\alpha}) = \frac{2}{3}\tilde{\alpha}_1$$

$$\Rightarrow \sqrt{n}\,\frac{g(\tilde{\alpha}) - g(\alpha)}{\sigma(\tilde{\alpha})} \to N(0,1);$$

$$\sigma^2(\alpha) = \nabla^T g(\alpha)\, K\, \nabla g(\alpha) = \frac{4}{9}(\alpha_2 - \alpha_1^2)$$

$$\Rightarrow \sqrt{n}\,\frac{\tilde{\theta}_1 - \theta}{\frac{2}{3}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} \to N(0,1) \Rightarrow t_1 = u_{\frac{1-\beta}{2}}, \quad t_2 = u_{\frac{1+\beta}{2}}$$

$$t_1 < \sqrt{n}\,\frac{\tilde{\theta}_1 - \theta}{\frac{2}{3}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} < t_2 \Rightarrow \tilde{\theta}_1 - \frac{2}{3}t_2\frac{\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{\sqrt{n}} < \theta < \tilde{\theta}_1 - \frac{2}{3}t_1\frac{\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{\sqrt{n}}$$

довер. интервал

Пункты f, g и h сделаны в коде

№ 76

$$p(x) = \begin{cases} \dfrac{\theta - 1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases} \qquad \theta > 1$$

а) <u>Оценка θ методом Макс. правдоподобия</u>

$$L(\theta) = \prod_{i=1}^{n} \left(\frac{\theta - 1}{x_i^{\theta}} \right) = (\theta - 1)^n \prod_{i=1}^{n} \frac{1}{x_i^{\theta}} = (\theta - 1)^n \left(\prod_{i=1}^{n} \frac{1}{x_i} \right)^{\theta} \Rightarrow$$

$$\Rightarrow \ln L(\theta) = n \ln(\theta - 1) + \theta \ln\left(\prod_{i=1}^{n} \frac{1}{x_i} \right) \Rightarrow \frac{\partial}{\partial \theta} \Rightarrow$$

$$\Rightarrow \frac{\partial}{\partial \theta} \ln L(\theta) = \frac{n}{\theta - 1} + \ln\left(\prod_{i=1}^{n} \frac{1}{x_i} \right) = 0 \Rightarrow$$

$$\Rightarrow \theta = 1 + \frac{n}{-\ln\left(\prod_{i=1}^{n} \frac{1}{x_i} \right)}$$ Берем $\frac{\partial^2}{\partial \theta^2}$ для опред т. max или т. min $\Rightarrow$

$$\Rightarrow \frac{\partial^2}{\partial \theta^2} (\ln L(\theta)) = -\frac{n}{(\theta - 1)^2} < 0 \Rightarrow$$ точка MAX

$$\Rightarrow \tilde{\theta} = 1 - \frac{n}{\ln\left(\prod_{i=1}^{n} \frac{1}{x_i} \right)}$$

б) <u>Довер. Интервал для Медианы</u>

$$\int_{-\infty}^{med} p(x)\,dx = \int_{-\infty}^{med} \frac{\theta - 1}{x^{\theta}}\,dx = (\theta - 1) \left(\frac{1}{(1 - \theta)\, med^{\theta - 1}} - \frac{1}{(1 - \theta) 1} \right) =$$

$$= 1 - \frac{1}{med^{\theta - 1}} = \frac{1}{2} \Rightarrow med = 2^{\frac{1}{\theta - 1}}$$

Нужно проверить на сильную регулярность.

для этого $F(x, \theta)$ <u>должна быть</u> регулярна, дважды непр. дифф на $(1; +\infty)$ и $\frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF(x, \theta) = \int_{-\infty}^{+\infty} d\, \frac{\partial^2 F(x, \theta)}{\partial \theta^2}$

непр на $(1;+\infty)$ => $F(x,\theta)$ - тоже дифф

$$\frac{\partial^2 F}{\partial \theta^2} = \frac{\partial^2}{\partial \theta^2}\left(1 - \frac{1}{x^{\theta-1}}\right) = \frac{\partial}{\partial \theta}\left(-\frac{\partial}{\partial \theta} x^{(1-\theta)}\right) = \frac{\partial}{\partial \theta}\Big(\underbrace{x^{(1-\theta)} \ln x}_{\frac{\partial F}{\partial \theta}}\Big) = \boxed{\frac{\ln^2 x}{x^{\theta-1}}}$$

непр на $(1;+\infty)$

=> $F(x,\theta)$ - дважды непр. дифф.! на $(1;+\infty)$!

$$\frac{\partial}{\partial \theta}\int_{-\infty}^{+\infty} dF = 0 \; ; \; \cancel{\frac{\partial}{\partial \theta}} \int_{-\infty}^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right) = \int_1^{+\infty} \frac{\partial p}{\partial \theta}\, dx =$$

$$= \int_1^{+\infty} \frac{x^{\theta} - (\theta-1)x^{\theta}\ln x}{x^{2\theta}}\, dx = \int_1^{+\infty} \frac{1}{x^{\theta}}\, dx - (\theta-1)\int_1^{+\infty} \frac{\ln x}{x^{\theta}}\, dx$$

$$\int_1^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right) = \frac{\ln x}{x^{\theta-1}}\Big|_1^{+\infty} = 0 - \underset{\ln x|_1 = 0}{0} = 0 \Rightarrow \boxed{\frac{\partial}{\partial \theta}\int_{-\infty}^{+\infty} dF = \int_{-\infty}^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right)}$$

$$\frac{\ln x}{x^{\theta-1}} \xrightarrow[x\to+\infty]{} 0, \quad \boxed{\theta > 1}$$

Проверим интегрируемость $I(\theta)$ для регулярности

$$I(\theta) = \int_{A(\theta)} \left(\frac{\partial \ln p}{\partial \theta}\right)^2 p\, dx = \int_1^{+\infty} \left(\frac{1}{\theta-1} - \ln x\right)^2 \frac{\theta-1}{x^{\theta}}\, dx =$$

$$= (\theta-1)\int_1^{+\infty} \left(\frac{1}{(\theta-1)^2} - \frac{2\ln x}{\theta-1} + \ln^2 x\right)\frac{1}{x^{\theta}}\, dx = \left\{\begin{array}{l} t\in[0;+\infty) \\ x = e^t \\ dx = e^t dt \end{array}\right\} =$$

$$= \underbrace{\int_0^{+\infty} \frac{1}{\theta-1} e^{-t(\theta-1)}\, dt}_{I_1} - 2\underbrace{\int_0^{+\infty} t e^{-(\theta-1)t}\, dt}_{I_2} + (\theta-1)\underbrace{\int_0^{+\infty} t^2 e^{-(\theta-1)t}\, dt}_{I_3}$$

$$I_1 = \frac{1}{(\theta-1)^2}\int_0^{+\infty} e^{-t(\theta-1)}\, d((\theta-1)t) = \frac{1}{(\theta-1)^2}\left(-e^{-(\theta-1)t}\right)\Big|_0^{+\infty} = \frac{1}{(\theta-1)^2}$$

тут → 0; тут $e^0 = 1$

$$I_2 = \int_0^{+\infty} t e^{-t(\theta-1)}\, \frac{d((\theta-1)t)}{\theta-1} = -\frac{1}{\theta-1}\int t\, d\left(e^{-t(\theta-1)}\right) =$$

$$= \frac{1}{\theta-1}\left(t e^{-(\theta-1)t}\Big|_0^{+\infty} + \int_0^{+\infty} e^{-t(\theta-1)}\, dt\right) = \frac{1}{(\theta-1)^2}\int_0^{+\infty} e^{-t(\theta-1)}\, d(t(\theta-1)) =$$

тут → 0; тут → 0; => = 0

$$= \frac{1}{(\theta-1)^2}\left(-e^{-(\theta-1)t}\right)\Big|_0^{+\infty} = \frac{1}{(\theta-1)^2}$$

тут → 0; тут = 1

$$I_3 = \int_0^{+\infty} t^2 e^{-t(\theta-1)}\, d(t(\theta-1)) = \underbrace{-t^2 e^{-t(\theta-1)}\Big|_{0}^{+\infty}}_{= 0} + \int_0^{+\infty} 2t e^{-t(\theta-1)}\, dt =$$

тут $\to 0$

тут $= 0$

$$= 2 I_2 = \frac{2}{(\theta-1)^2}$$

Тогда: $I(\theta) = \frac{1}{(\theta-1)^2}$ — непрерывна на любом компакте из $(1; +\infty)$

$\Rightarrow$ Т.к. $I(\theta)$ — непр. на любом компакте из $(1; +\infty)$

и $\frac{\partial}{\partial\theta}\int_{-\infty}^{+\infty} dF(x,\theta) = \int_{-\infty}^{+\infty} d\,\frac{\partial F}{\partial\theta}$

$\Rightarrow$ Модель Регулярна

$$\frac{\partial^2}{\partial\theta^2}\int_{-\infty}^{+\infty} dF = 0 \quad ; \quad \int_{-\infty}^{+\infty} d\left(\frac{\partial^2 F}{\partial\theta^2}\right) = \int_1^{+\infty} d\left(\frac{\ln^2 x}{x^{\theta-1}}\right) = \frac{\ln^2 x}{x^{\theta-1}}\Big|_1^{+\infty} = 0$$

тут $\to 0$

тут $= 0$

$$\Rightarrow \frac{\partial^2}{\partial\theta^2}\int_{-\infty}^{+\infty} dF = 0 = \int_{-\infty}^{+\infty} d\left(\frac{\partial^2 F}{\partial\theta^2}\right) \quad *$$

$\Rightarrow$ Из регулярности, дважды непр. дифф. $F(x,\theta)$ на $(1; +\infty)$ $\forall\theta$

и $*$ $\Rightarrow$, что Модель Сильно Регулярна

$$\Rightarrow \sqrt{n}(\tilde\theta - \theta) \rightsquigarrow N\left(0, \frac{1}{I(\theta)}\right)$$

$med = 2^{\frac{1}{x-1}}$ — дифф ; $med'\theta = 2^{\frac{1}{x-1}} \ln 2 \,\frac{-1}{(x-1)^2}\Big|_{x=\theta} = -2^{\frac{1}{\theta-1}} \ln 2 \cdot \frac{1}{(\theta-1)^2} \neq 0$ при $\theta > 1$

$\Rightarrow$ Работает $\sqrt{n}\,(med(\tilde\theta) - med(\theta)) \rightsquigarrow N\left(0, med'^2(\theta)\,\frac{1}{I(\theta)}\right)$

$$\sigma(\theta) = 2^{\frac{1}{\theta-1}} \ln 2 \,\frac{1}{(\theta-1)^2}\,\cancel{(\theta-1)} = \frac{\ln 2 \;\, 2^{\frac{1}{\theta-1}}}{\theta-1}$$

$$\Rightarrow \sqrt{n}\,\frac{med(\tilde\theta) - med(\theta)}{\sigma(\tilde\theta)} \rightsquigarrow N(0; 1)$$

$t_1 = u_{\frac{1-\beta}{2}}$ $\quad t_2 = u_{\frac{1+\beta}{2}}$

$$t_1 < \sqrt{n}\,\frac{(med(\tilde{\theta}) - med(\theta))}{\sigma(\tilde{\theta})} < t_2 \Rightarrow$$

$$\Rightarrow med(\tilde{\theta}) - \frac{\sigma(\tilde{\theta})}{\sqrt{n}} t_2 < med(\theta) < med(\tilde{\theta}) - \frac{\sigma(\tilde{\theta})}{\sqrt{n}} t_1$$

c) <u>Асимптотический доверительный интервал для θ</u>

$$\sqrt{n}(\tilde{\theta} - \theta) \sim N\left(0; \frac{1}{I(\theta)}\right) \Rightarrow$$

$$\Rightarrow \sqrt{n}(\tilde{\theta} - \theta)\sqrt{I(\tilde{\theta})} \sim N(0, 1) \Rightarrow$$

$$\Rightarrow \tilde{\theta} - \frac{t_2}{\sqrt{n I(\tilde{\theta})}} < \theta < \tilde{\theta} - \frac{t_1}{\sqrt{n I(\tilde{\theta})}}$$

Пункты d, e, f – в коде.

Все же напишу вывод тут. Я взял $\theta = 17$

Асс. для med: (1,0395; 1,0594)

Асс. для θ: (12.546; 18.175) $\quad l = 5.629$

Неп. Bootstrap: (12.292; 17.449) $\quad l = 5.157$

Пар. бутстрап: (12.029; 17.813) $\quad l = 5.784$

А так же вывод кода для $\boxed{№5}$

Точн. метод (9.998; 10.181) $\quad l = 0.183$

Ас. метод для (10.015; 10.523) $\quad l = 0.508$

Неп. Bootstrap (9.883; 10.634) $\quad l = 0.751$